

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5303
Subject Title	Robust Control Technology in Low-Altitude Aerial Vehicle
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to: 1. introduce the principles, concepts and models of most popular spatial perception algorithms for low-altitude aerial vehicles. 2. provide both theoretical and practical understanding of spatial perception models for low-altitude aerial vehicles such as machine learning and deep learning. 3. develop proficiency in designing, training, and optimizing spatial perception models for low-altitude aerial vehicles using modern programming language.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand the fundamental principles and algorithms of spatial perception information, data and models for low-altitude aerial vehicles; b. develop insights into the effectiveness and applicability of spatial perception techniques in the low-altitude industry; c. gain knowledge of applying spatial perception techniques for aerial vehicles to cutting-edge low-altitude applications; d. design innovative spatial perception solutions to address complex challenges in low-altitude perception, considering various requirements and constraints; and e. participate effectively in teamwork, presentation, and technical writing, enhancing collaboration and communication skills.
Subject Synopsis/ Indicative Syllabus	Advanced Sensing Technologies for Low-Altitude Aerial Vehicles: Overview of Sensory Technology: Detailed study of various sensory technologies such as GNSS, LiDAR, visual sensors, mmWave radars, thermal cameras, and depth sensors, emphasizing their role in enhancing vehicle navigation and obstacle detection. Operational Integration: Techniques for the integration and calibration of these instruments to achieve optimal functionality in various environmental conditions, including urban and rural landscapes. Data Integration and Analysis Techniques for Aerial Systems: Data Handling Techniques: Exploration of methods for managing large volumes of data from multiple sensors, focusing on synchronization,

	alignment, and real-time processing challenges. Advanced Analysis: Use of sophisticated algorithms for noise reduction, feature extraction, and the combination of data sources to create comprehensive environmental models, enhancing decision-making processes. Computational Tools and Methodologies for Aerial Systems: Toolset Proficiency: In-depth training in the use of modern programming language for developing and testing machine learning models. Introduction to software environments and libraries specifically suited for aerial data analysis. Learning Techniques: Detailed examination of supervised, unsupervised, and reinforcement learning paradigms and their applicability to tasks such as predictive modeling, anomaly detection, and adaptive perception in low-altitude contexts. Robust Scene Perception in Dynamic Environments: Scene Analysis: Comprehensive coverage of algorithms for dynamic scene understanding, including visual odometry for tracking vehicle movement and 3D mapping to create detailed environmental representations. Algorithmic Challenges: Discussion of the challenges in implementing these algorithms in low-altitude scenarios, such as dealing with variable lighting conditions, weather impacts, and moving obstacles. Practical Applications in Low-Altitude Aerial Systems: Data-Driven Solutions: Focuses on applying techniques such as pattern discovery, topic modeling, and predictive analytics to enhance navigation and operational decisions in low-altitude environments. Practical Considerations: Discusses scalability and interpretability of machine learning models that are crucial for real-time applications in low-altitude systems. Also, addresses legal, social, and ethical considerations in deploying data analytics in this context.																																			
Teaching/Learning Methodology	The course employs a combination of lectures, tutorials, lab sessions, and a group project to facilitate learning. Lectures provide comprehensive coverage of the course concepts, supplemented by examples and interactive question & answer sessions for clarity. Tutorials and lab sessions reinforce theoretical knowledge with practical exercises, emphasizing hands-on experience with robust spatial perception tools and techniques. The group project engages students in collaborative work to apply theoretical concepts to real-world scenarios, enhancing their analytical and problem-solving skills. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Lab</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Project</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	e	Lecture	√	√	√	√		Tutorial	√	√	√	√		Lab	√	√	√	√	√	Project	√	√	√	√	√
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Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	1. Assignments	20%	√	√	√	√	
	2. Quizzes	30%	√	√	√	√	
	3. Project Development	50%	√	√	√	√	√
	Total	100 %					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 1 × Continuous Assessment Continuous assessments in this course include assignments, quizzes, application development, project reports, and presentations, each structured to achieve intended learning outcomes centered around spatial perception for low-altitude aerial vehicles. Assignments are designed to deepen students' understanding of robust spatial perception models and algorithms, offering practical experience with software tools to foster insights into their effectiveness in aerial applications. Quizzes confirm students' mastery of core concepts and their application to innovative percpetion systems for low-altitude vehicles. Project Development includes application development, project report and project presentation. Application development focuses on leveraging spatial perception and robust learning principles to tackle real-world challenges specific to low- altitude aerial vehicles, assessing students' capability to engineer comprehensive solutions that address unique operational demands. Project reports and presentations allow students to demonstrate their analytical and problem-solving skills by implementing, analyzing, and communicating their projects, ensuring a thorough understanding of integrating robust spatial perception tailored to the aerial vehicle context. This cumulative assessment strategy ensures that students develop a solid foundation in the theoretical and practical aspects of spatial perception applied specifically to low-altitude aerial vehicles.						
Student Study Effort Expected	Class contact:						
	▪ Lecture/Tests					26 Hrs.	
	▪ Laboratory/Tutorial					13 Hrs.	
	Other student study effort:						
	▪ Assignments, Projects, Quizzes					36 Hrs.	
	▪ Literature Review, Self-study & Preparation					36 Hrs.	
	Total student study effort					111 Hrs.	

Reading List and References	1. Zhang A, Lipton ZC, Li M, Smola AJ. Dive into deep learning. Cambridge University Press, 2023. 2. R. Siegwart, I.R. Nourbakhsh, and D. Scaramuzza. Introduction to autonomous mobile robots 2nd Edition, MIT, 2011. 3. EMC Education Services (Editor), Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data, Wiley, 2015. 4. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2006. 5. Deisenroth MP, Faisal AA, Ong CS. Mathematics for machine learning. Cambridge University Press; 2020 Apr 23. 6. Hartley, Richard, and Andrew Zisserman. Multiple view geometry in computer vision. Cambridge university press, 2003. 7. D2L.ai: Interactive Deep Learning Book with Multi-Framework Code, Math, and Discussions. https://github.com/d2l-ai/d2l-en?tab=readme-ov-file. 2023
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Sep 2024